

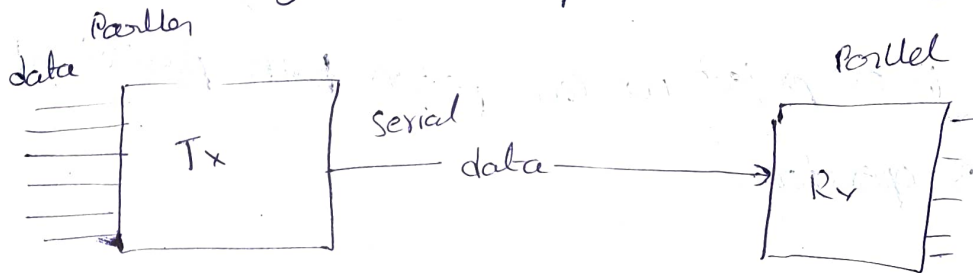
UART RECEIVER / Transmitter:

Description:

- > UART (Universal Asynchronous Receiver / Transmitter) is a serial communication system which converts parallel data into serial while transmission.
- > It can be used for serial data also.

DEF:

"UART is a serial communication system used to transfer digital data between two devices without using a separate clock line."



Ex: 8 bit UART

data = 10101010

data frame

Idle	start	data	stop
1	0	10101010	1
LSB			MSB

Idle = 1 → transmitter create serial bit.

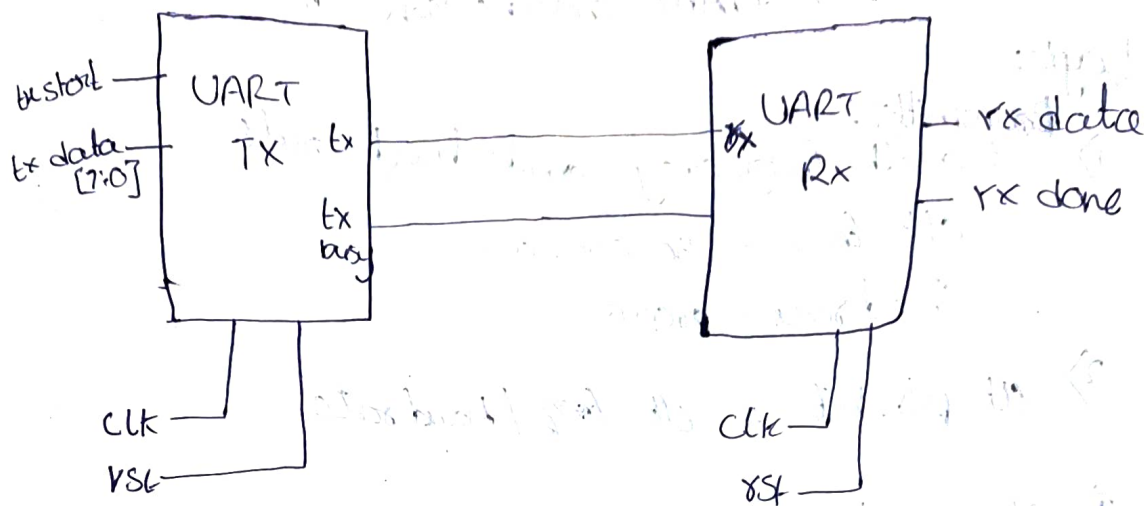
At receiver

Rx = 0 then receiver starts

Rx = 1 receiving ends

start	data	stop
0	10101010	1
		MSB

Block diagram:



Ports:

clk, rst $\rightarrow$ Tx, Rx has different clk and rst source
because UART is an Asynchronous system.

tx_start - Input data port of Tx

$\rightarrow$ If tx_start = 0 $\rightarrow$ don't transmit / stop transmitting
tx_start = 1 $\rightarrow$ start transmission

tx_data - Input port that transmits data

tx - output port that transmits data to Receiver
by converting parallel data to serial or serial
itself.

tx_busy - 0 $\rightarrow$ transmission completed

1 $\rightarrow$ transmission in progress

Rx $\rightarrow$ Input port of Receiver that receives data
in the form of serial data

rx_data $\rightarrow$ output port that transmits serial data in
parallel or serial.

Rx done - 0 $\rightarrow$ UART transmission in progress.

1 $\rightarrow$ UART transmission is done.

Logic:

Transmitter:

1) Set a clk frequency and baud rate

$$\text{CLK freq} = 50,000,000$$

$$\text{Baud rate} = 9600$$

2) $\text{clk_per_bit} = \text{clk freq} / \text{baud rate}$

3) At positive edge of clk or rs

if $\text{rst} = 1$: $\rightarrow$ get all ports to initial position

tx $\rightarrow 1$ tx-busy $\rightarrow 0$

clk_cnt $\rightarrow 0$ bit_index $\rightarrow 0$

$\hookrightarrow$ count no. of clocks $\hookrightarrow$ that denote every bit in input data

else

If tx_start = 1 and tx-busy = 0

tx-busy $\rightarrow 1$

clk_cnt $\rightarrow 0$ bit_index $\rightarrow 0$

reset the communication $\rightarrow$ data transmission is completed reset to initial position.

else tx-bus ! = 0

data transmission is in still progress.

If $(\text{clk_cnt} - \text{clk_per_bit})$

$$\text{clk_cnt} = \text{clk_count} + 1;$$

else begin

clk_cnt = 0

case (bit_index) [8:0] → decimal

u'd0: tx <= 1'b0 // Starting bit

u'd1: tx <= tx_data[0]

u'd8: tx <= tx_data[7];

u'd9: begin

tx <= 1'b1; // Stop bit

tx_busy <= 1'b0;

Receiver:

u → state

IDLE

STAT

DATA

STOP

at posedge of clk or rst

at rst

If rst = 1

reset every port to 0 and state to Idle

else

rx_done = 0 → means transmission in progress

case -

Idle → reset every port to zero

and state → start

Start $\rightarrow$

clk_counts ++ up-to $\text{clk_per_bit} / 2$

State $\leftrightarrow$ Idle if $rx = 1$

State $\rightarrow$ data if $rx = 0$

Data $\rightarrow$

at $\text{bit_index} = 0 \rightarrow$ start bit

$\text{bit_index} = 1 \rightarrow \text{rx_data}[c]$

$\text{bit_index} = 9 \rightarrow$ stop bit $\rightarrow$ check $rx \rightarrow$